

Empty slide

# Fractals

VP Summer Camp  
May 2026

Akshay Sanjeev K.





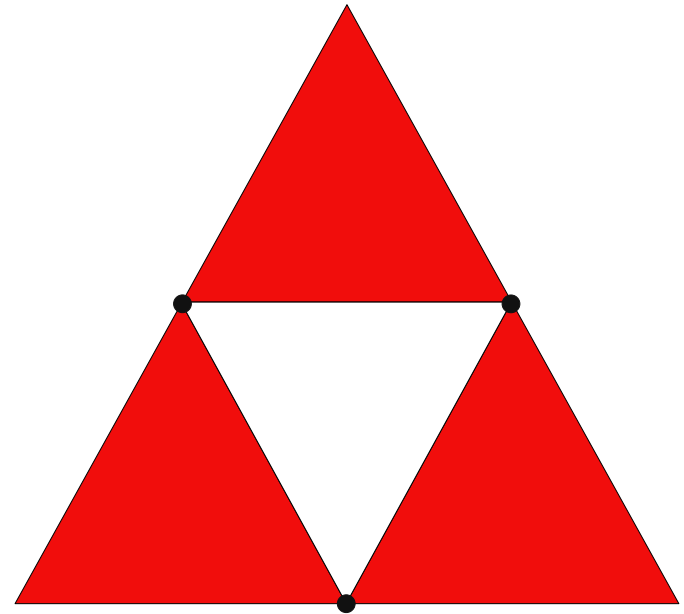
# Fractals are everywhere



# Sierpiński Triangle

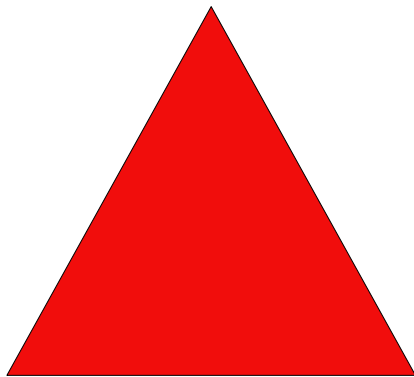


1. Consider a triangle.
2. Divide each side into half.
3. Connect the midpoints.
4. Remove the middle triangle.
5. Repeat step 1 to 4 iteratively.

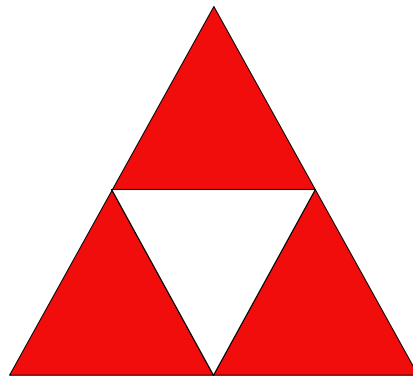


# Investigation #1

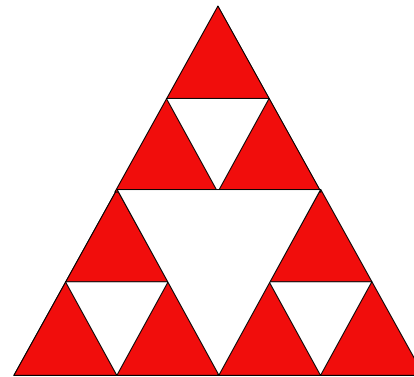
How many coloured triangles are there?



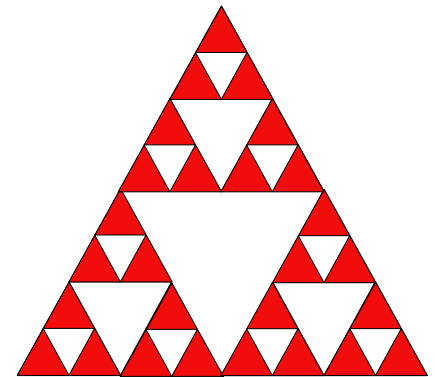
$$T(S_0) = 1$$



$$T(S_1) = 3$$



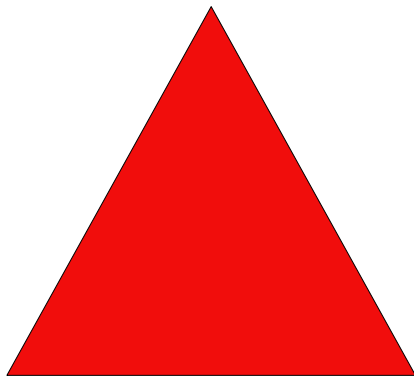
$$T(S_2) = 9$$



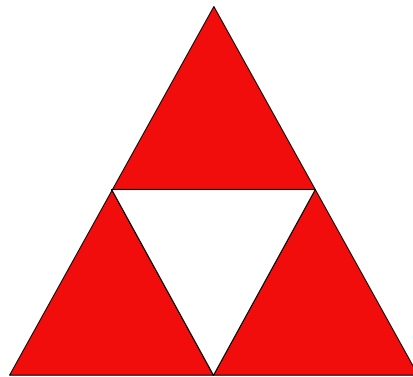
$$T(S_3) = 27$$

- How many shaded triangles are there in  $S_4$  ?
- Can you express  $T(S_4)$  in terms of  $T(S_3)$  ?
- How about a general expression for  $T(S_n)$  ?

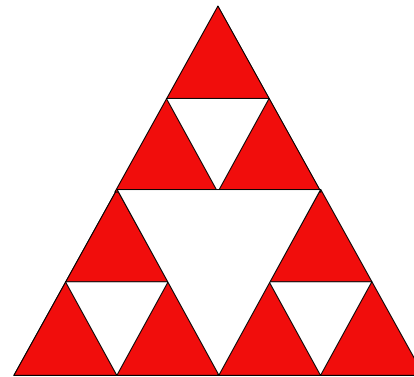
# Investigation #2



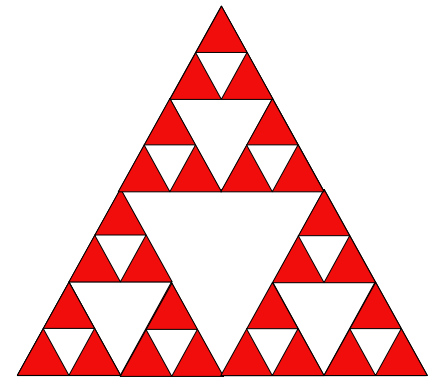
$$T(S_0) = 1$$



$$T(S_1) = 3$$



$$T(S_2) = 9$$



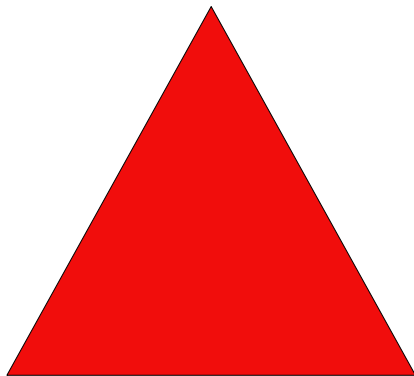
$$T(S_3) = 27$$

- How many copies of  $S_1$  do you find in  $S_2$  ?
- How many copies of  $S_2$  and  $S_1$  do you find in  $S_3$
- How many copies of  $S_{n-1}$ ,  $S_{n-2}, \dots, S_1$  will you find in  $S_n$

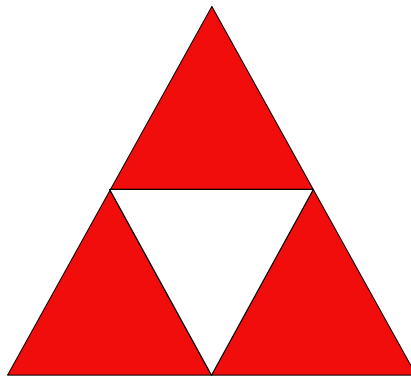


# Investigation #3

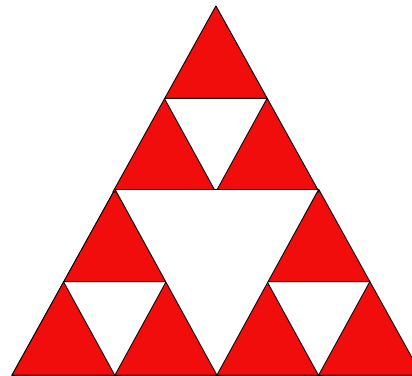
What is the area of coloured triangles ?



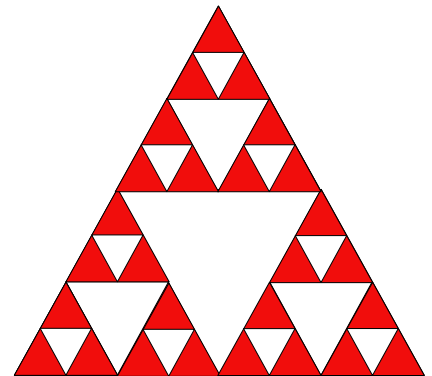
$$\text{Ar}(S_0) = A$$



$$\text{Ar}(S_1) = 3A/4$$



$$\text{Ar}(S_2) = 9A/16$$

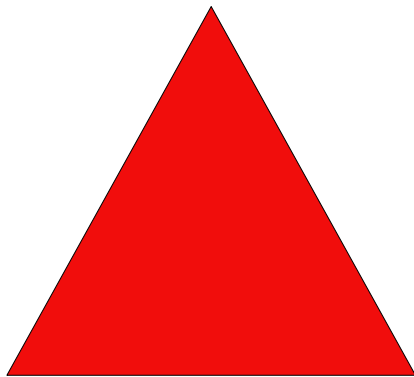


$$\text{Ar}(S_3) = 27A/64$$

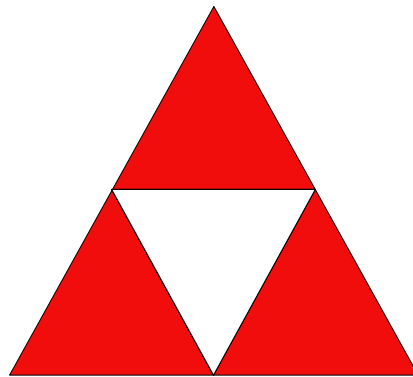
- Do you see any pattern emerging?
- What happens if we keep increasing  $n$ ?

# Investigation #4

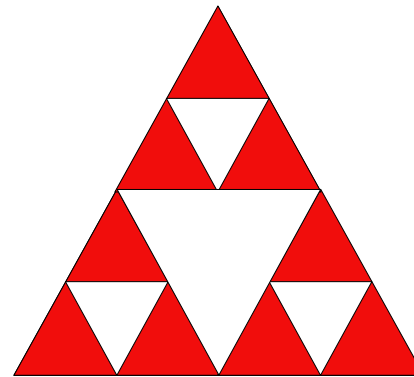
What is the length of boundary ?



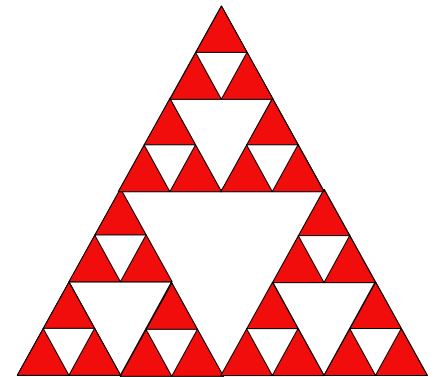
$$L(S_0) = 3$$



$$L(S_1) = 9/2$$



$$L(S_2) = 27/4$$

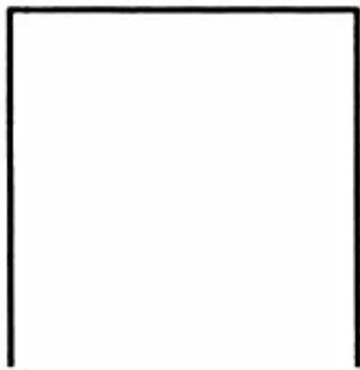


$$L(S_3) = 81/8$$

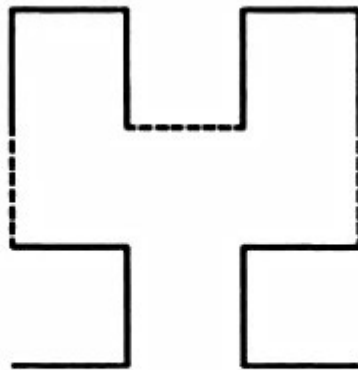
- Do you see any pattern emerging?
- What happens if we keep increasing  $n$ ?



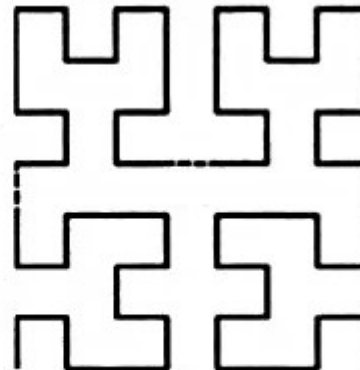
# More fractals



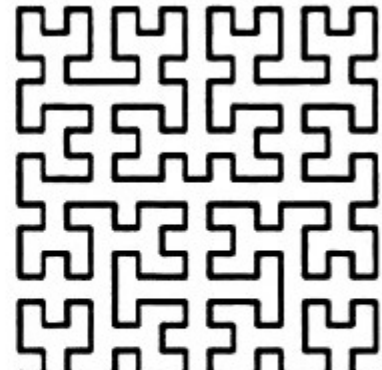
order=1



order=2



order=3



order=4

# Koch Snowflake

# Mandelbrot set